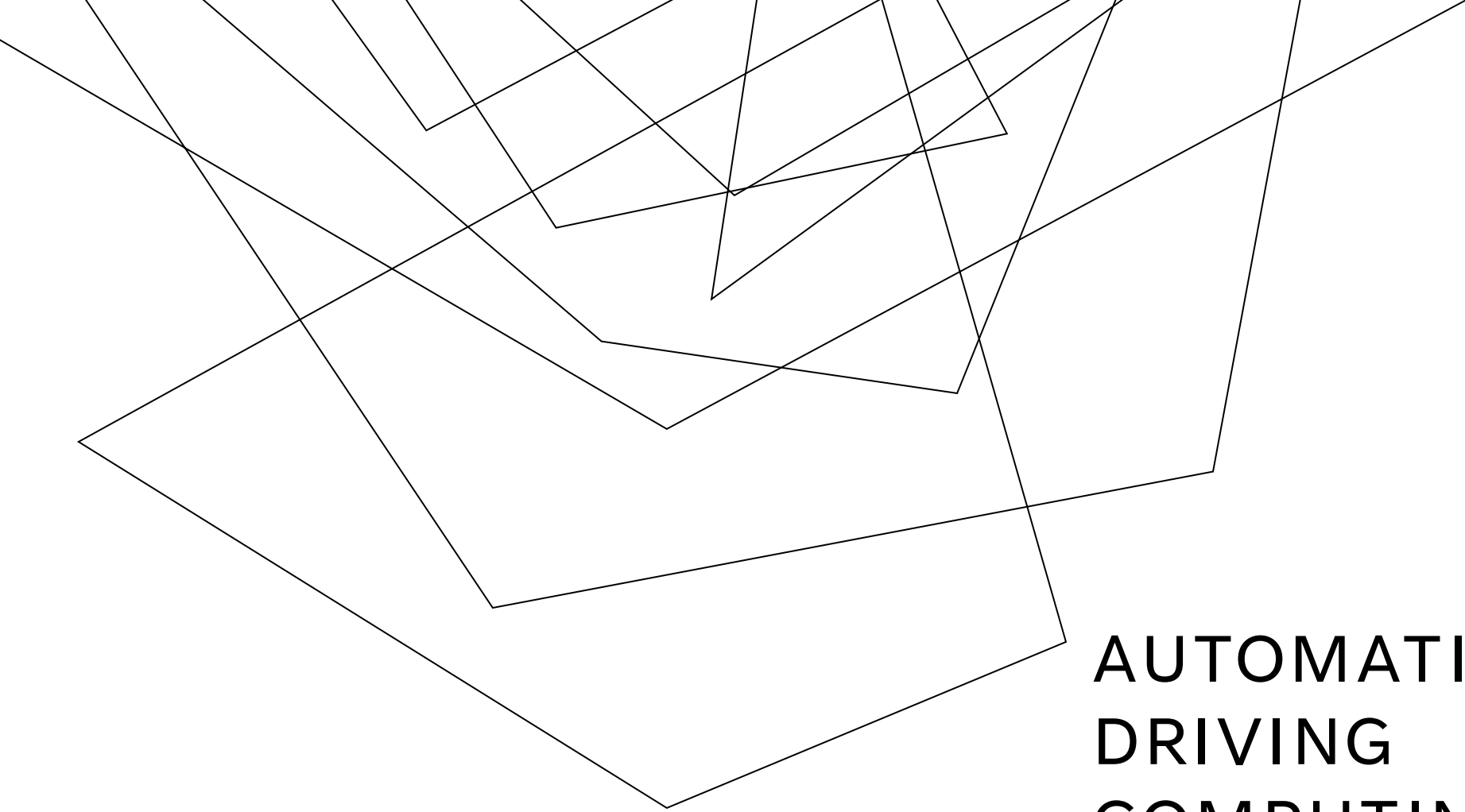


A series of thin, black, overlapping lines forming various geometric shapes like triangles and polygons, creating a complex, abstract pattern in the upper left portion of the slide.

AUTOMATIC DRIVING COMPUTING

8. PERCEPTION (6)





지난 시간 복습

OBJECT DETECTION (4)

1. Single-stage object detector





목차

목차

1. 2D Semantic / instance segmentation





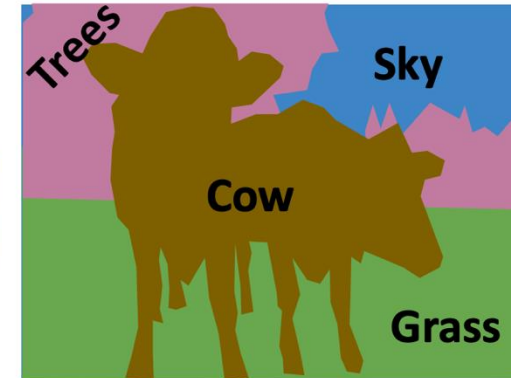
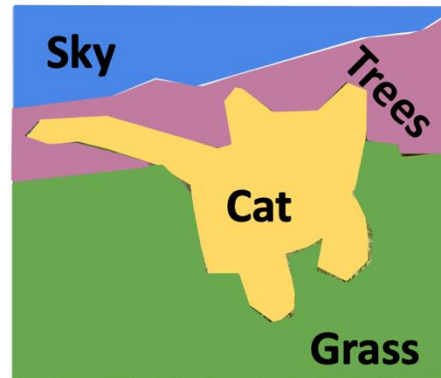
2D SEMANTIC / INSTANCE SEGMENTATION

SEMANTIC SEGMENTATION

Task definition : 이미지 내 모든 픽셀에 대해 카테고리 라벨을 assign
“pixel-level” task

Label each pixel in the image
with a category label

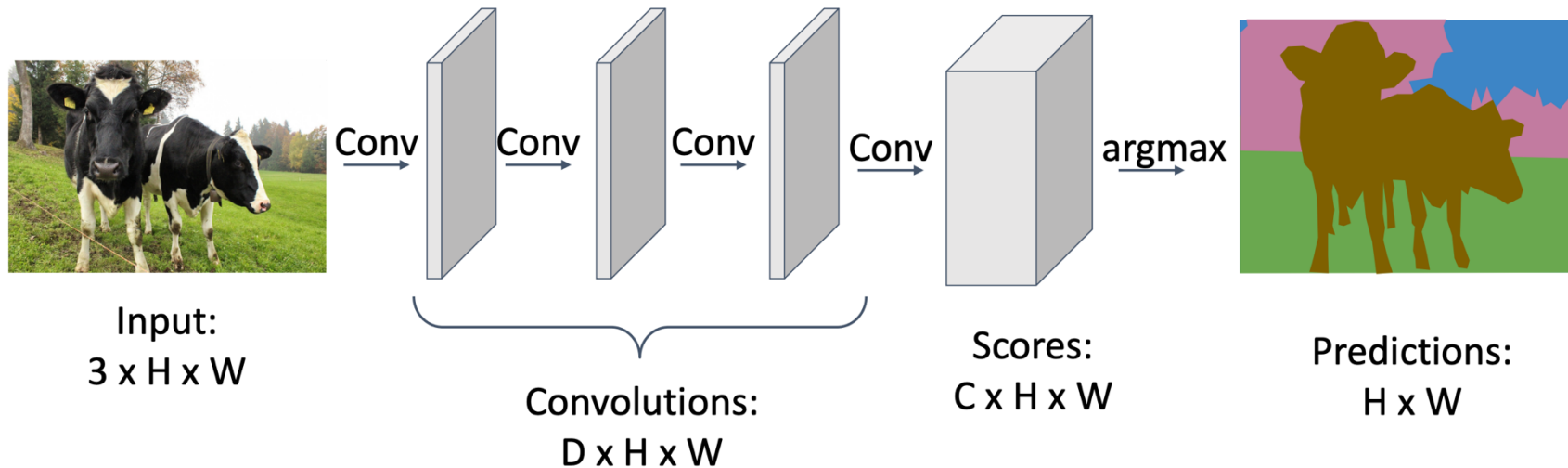
Don't differentiate instances,
only care about pixels



SEMANTIC SEGMENTATION

Semantic Segmentation: Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Loss function: Per-Pixel cross-entropy

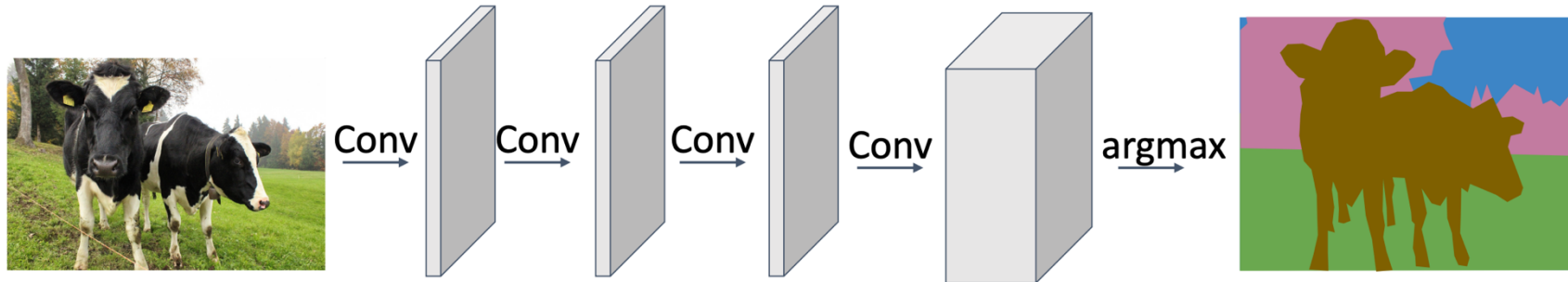
Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015



SEMANTIC SEGMENTATION

Semantic Segmentation: Fully Convolutional Network

Design a network as a bunch of convolutional layers to make predictions for pixels all at once!



Input:
 $3 \times H \times W$

Problem #1: Effective receptive field size is linear in number of conv layers: With L 3×3 conv layers, receptive field is $1+2L$

Problem #2: Convolution on high res images is expensive! Recall ResNet stem aggressively downsamples

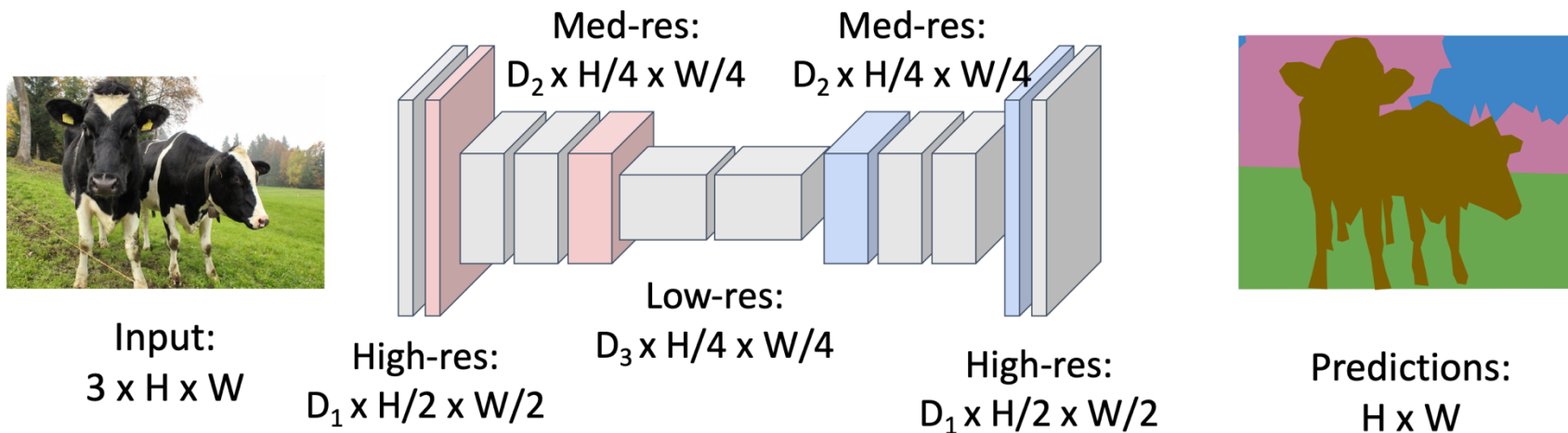
Long et al, "Fully convolutional networks for semantic segmentation", CVPR 2015



SEMANTIC SEGMENTATION

Semantic Segmentation: Fully Convolutional Network

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Long, Shelhamer, and Darrell, "Fully Convolutional Networks for Semantic Segmentation", CVPR 2015
Noh et al, "Learning Deconvolution Network for Semantic Segmentation", ICCV 2015



SEMANTIC SEGMENTATION

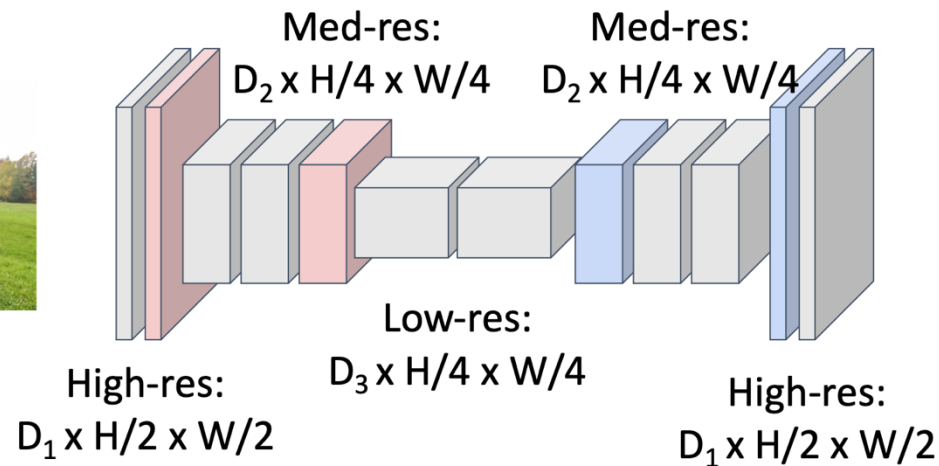
Semantic Segmentation: Fully Convolutional Network

Downsampling:
Pooling, strided
convolution



Input:
 $3 \times H \times W$

Design network as a bunch of convolutional layers, with **downsampling** and **upsampling** inside the network!



Upsampling:
???



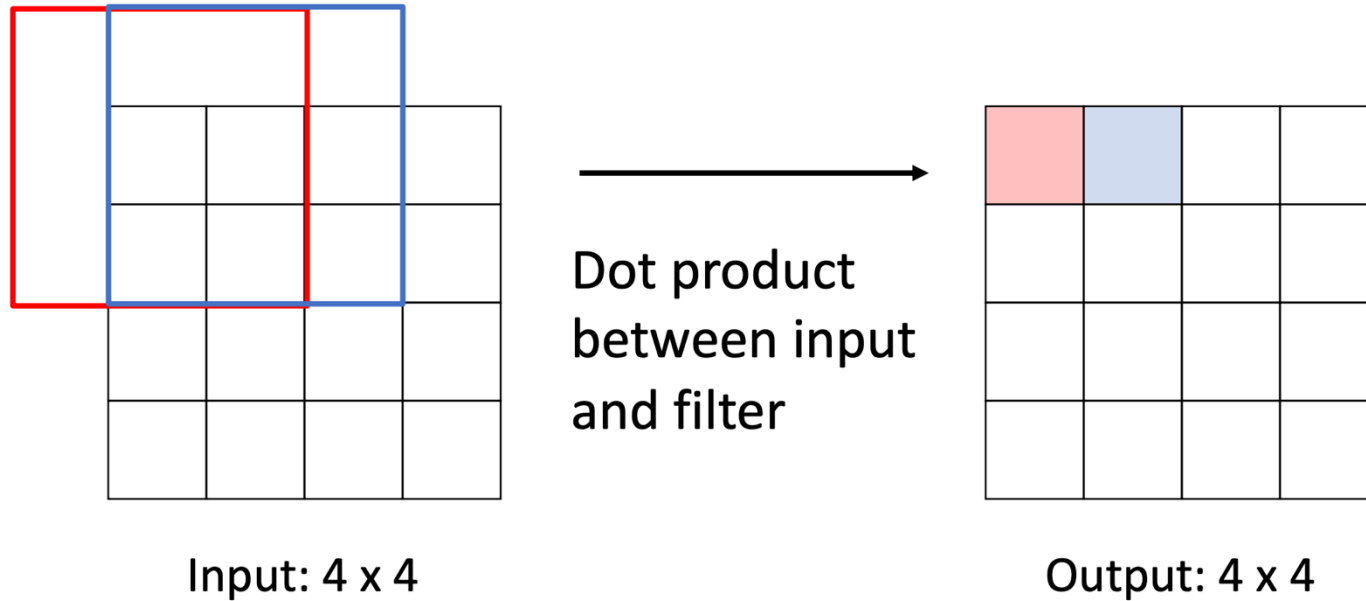
Predictions:
 $H \times W$



SEMANTIC SEGMENTATION

Learnable Upsampling: Transposed Convolution

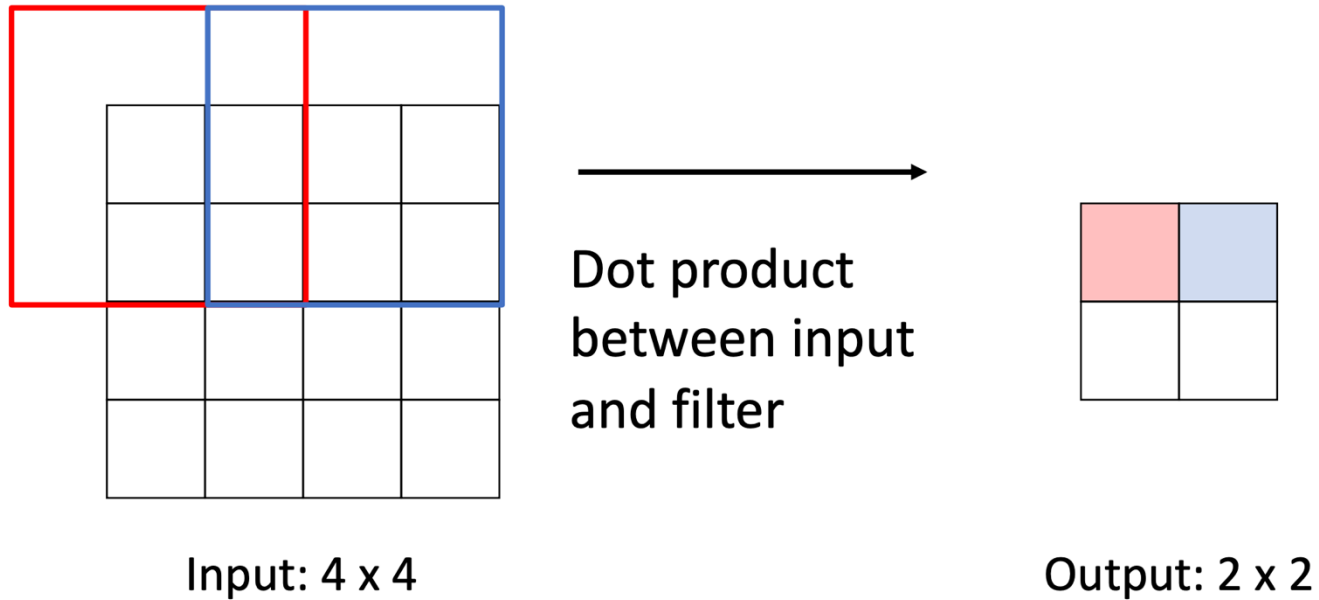
Recall: Normal 3 x 3 convolution, stride 1, pad 1



SEMANTIC SEGMENTATION

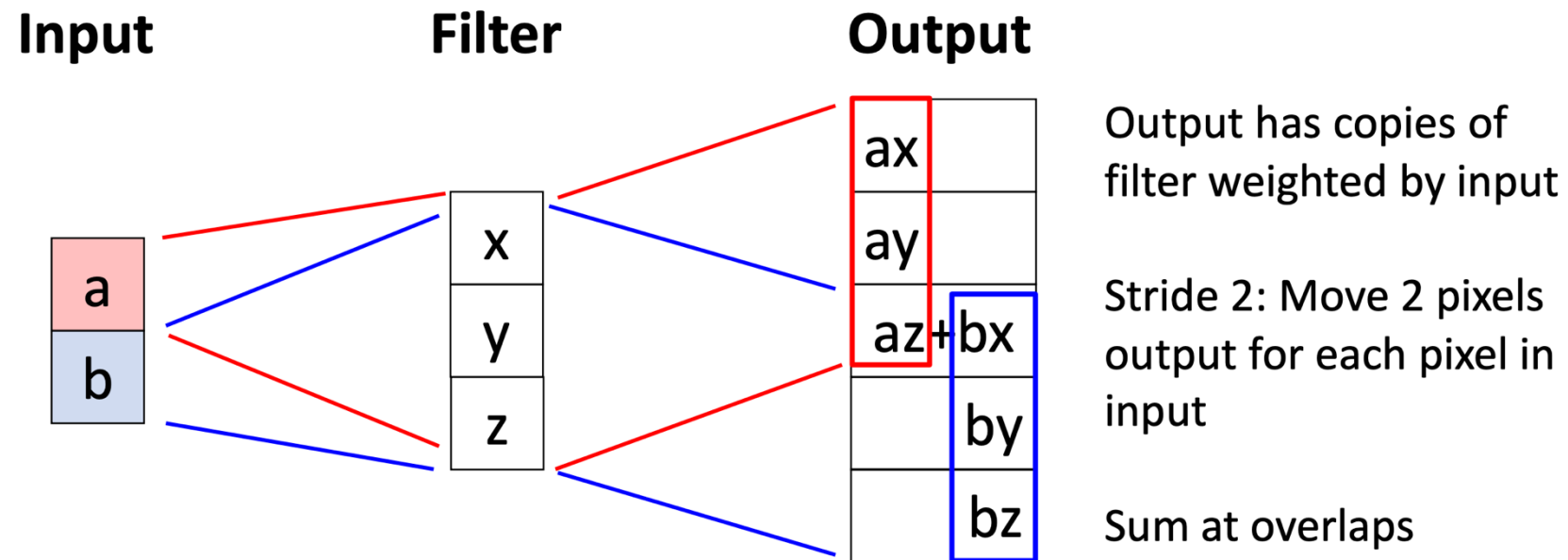
Learnable Upsampling: Transposed Convolution

Recall: Normal 3 x 3 convolution, stride 2, pad 1



SEMANTIC SEGMENTATION

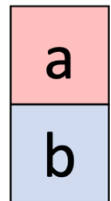
Transposed Convolution: 1D example



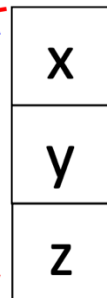
SEMANTIC SEGMENTATION

Transposed Convolution: 1D example

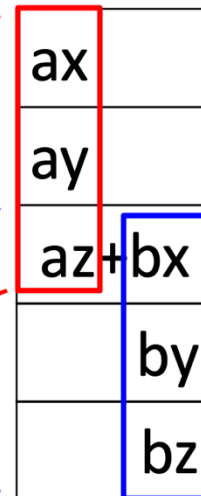
Input



Filter



Output



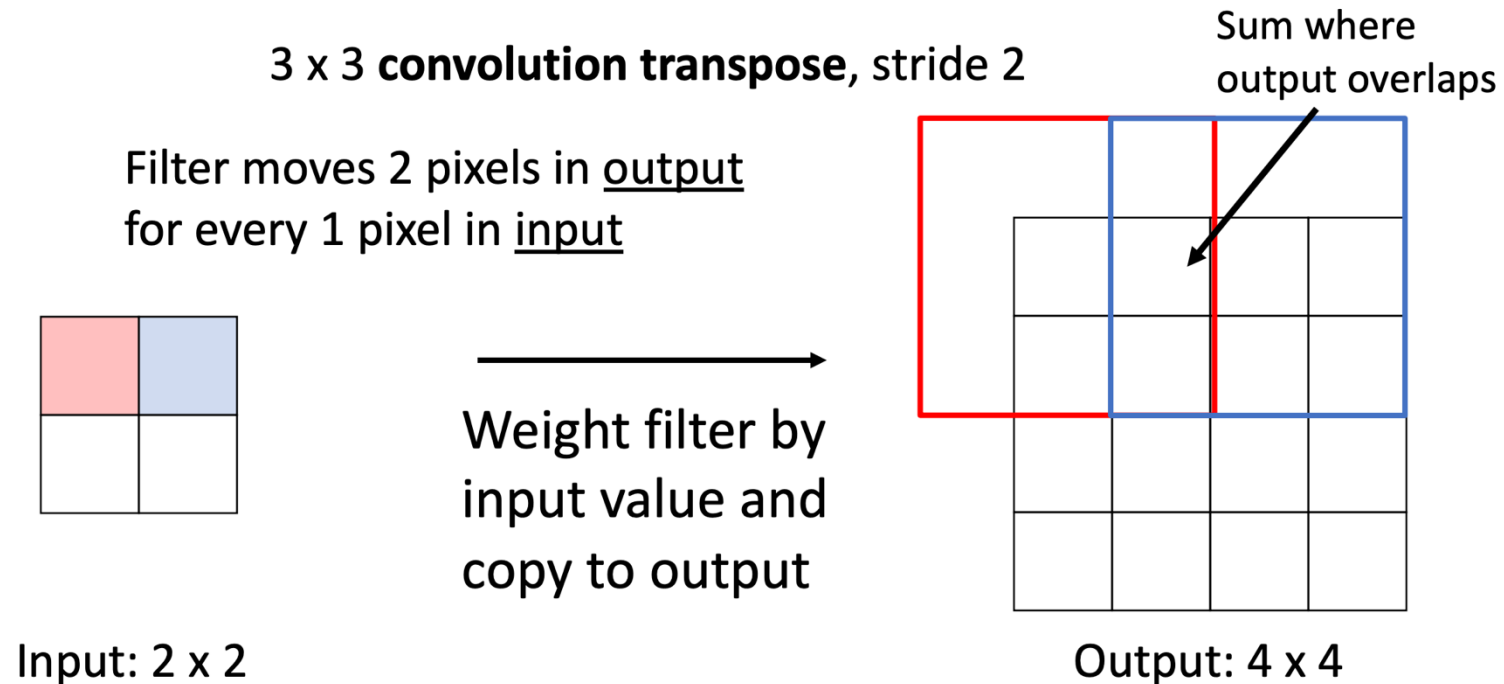
This has many names:

- Deconvolution (bad)!
- Upconvolution
- Fractionally strided convolution
- Backward strided convolution
- Transposed Convolution (best name)



SEMANTIC SEGMENTATION

Learnable Upsampling: Transposed Convolution

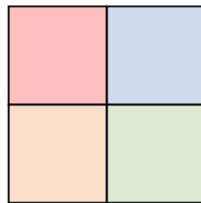


SEMANTIC SEGMENTATION

Learnable Upsampling: Transposed Convolution

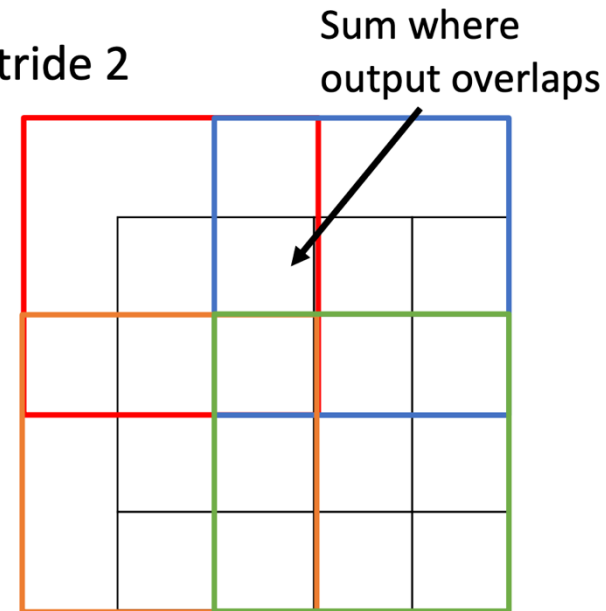
3 x 3 **convolution transpose**, stride 2

This gives 5x5 output – need to trim one pixel from top and left to give 4x4 output



Input: 2 x 2

→
Weight filter by
input value and
copy to output



Output: 4 x 4

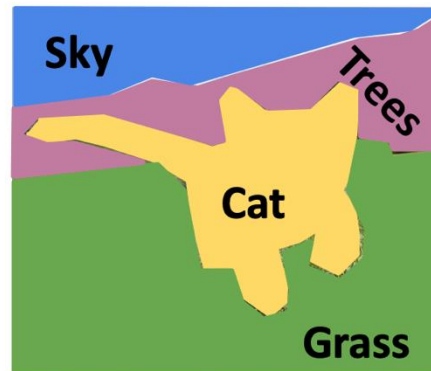


THINGS AND STUFF

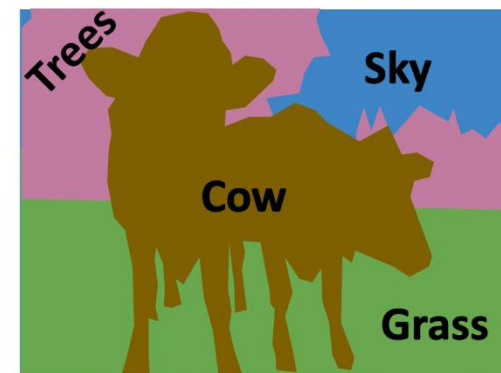
Things: Object categories that can be separated into object instances (e.g. cats, cars, person)



Stuff: Object categories that cannot be separated into instances (e.g. sky, grass, water, trees)



[This image is CC0 public domain](#)

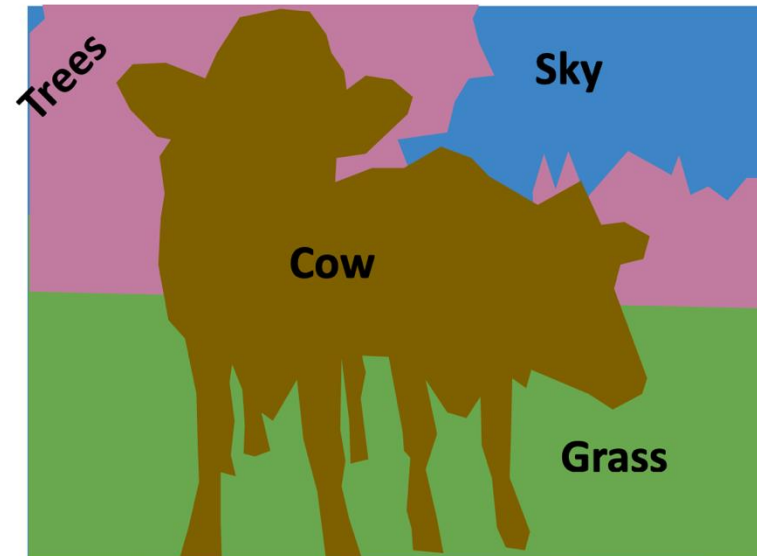


THINGS AND STUFF

Object Detection: Detects individual object instances, but only gives box (Only things!)



Semantic Segmentation: Gives per-pixel labels, but merges instances (Both things and stuff)

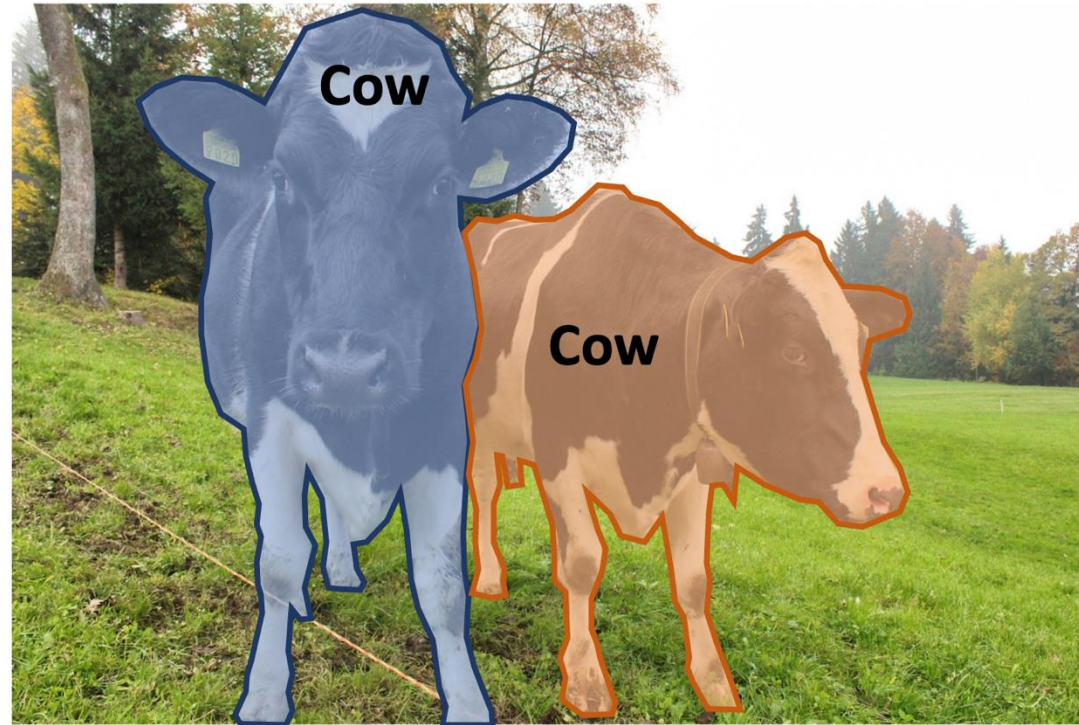


INSTANCE SEGMENTATION

Instance Segmentation:

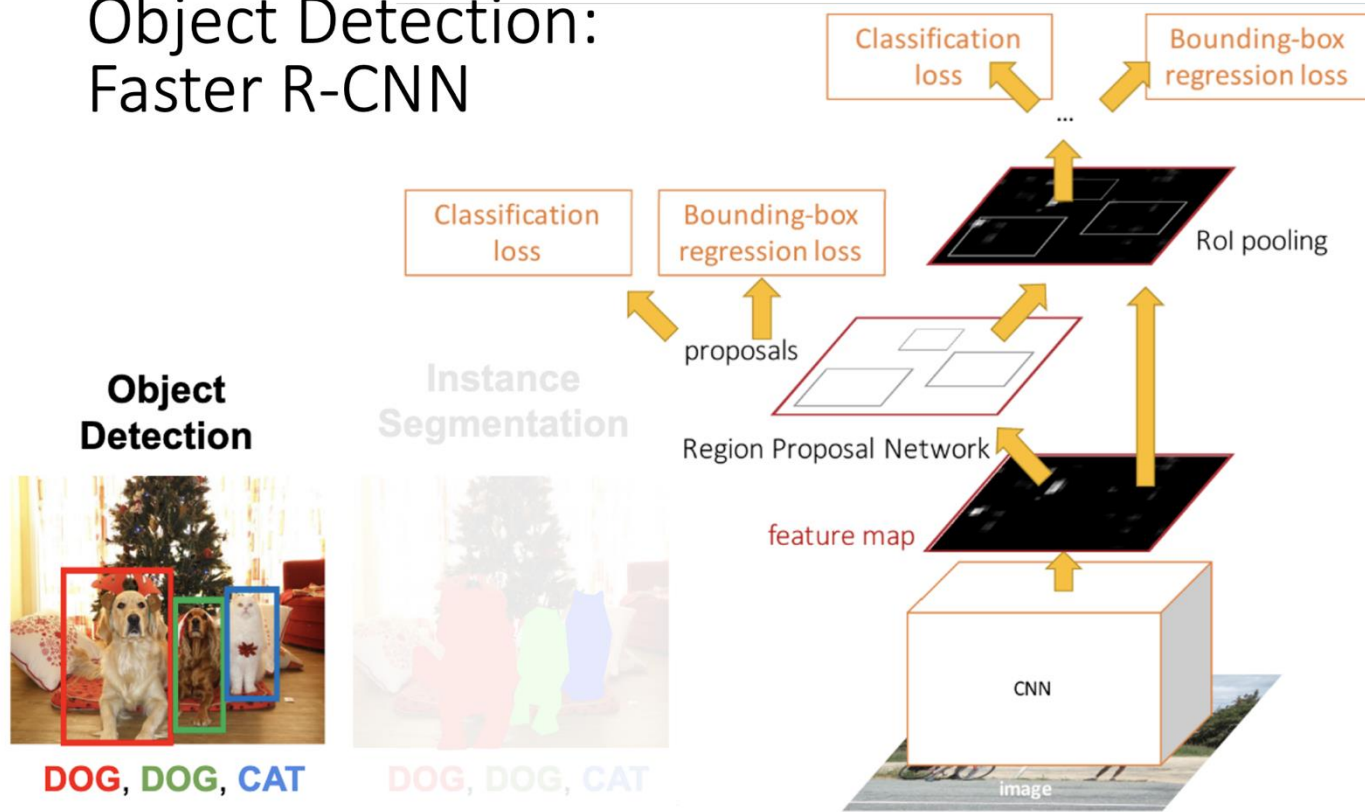
Detect all objects in the image, and identify the pixels that belong to each object (Only things!)

Approach: Perform object detection, then predict a segmentation mask for each object!



INSTANCE SEGMENTATION

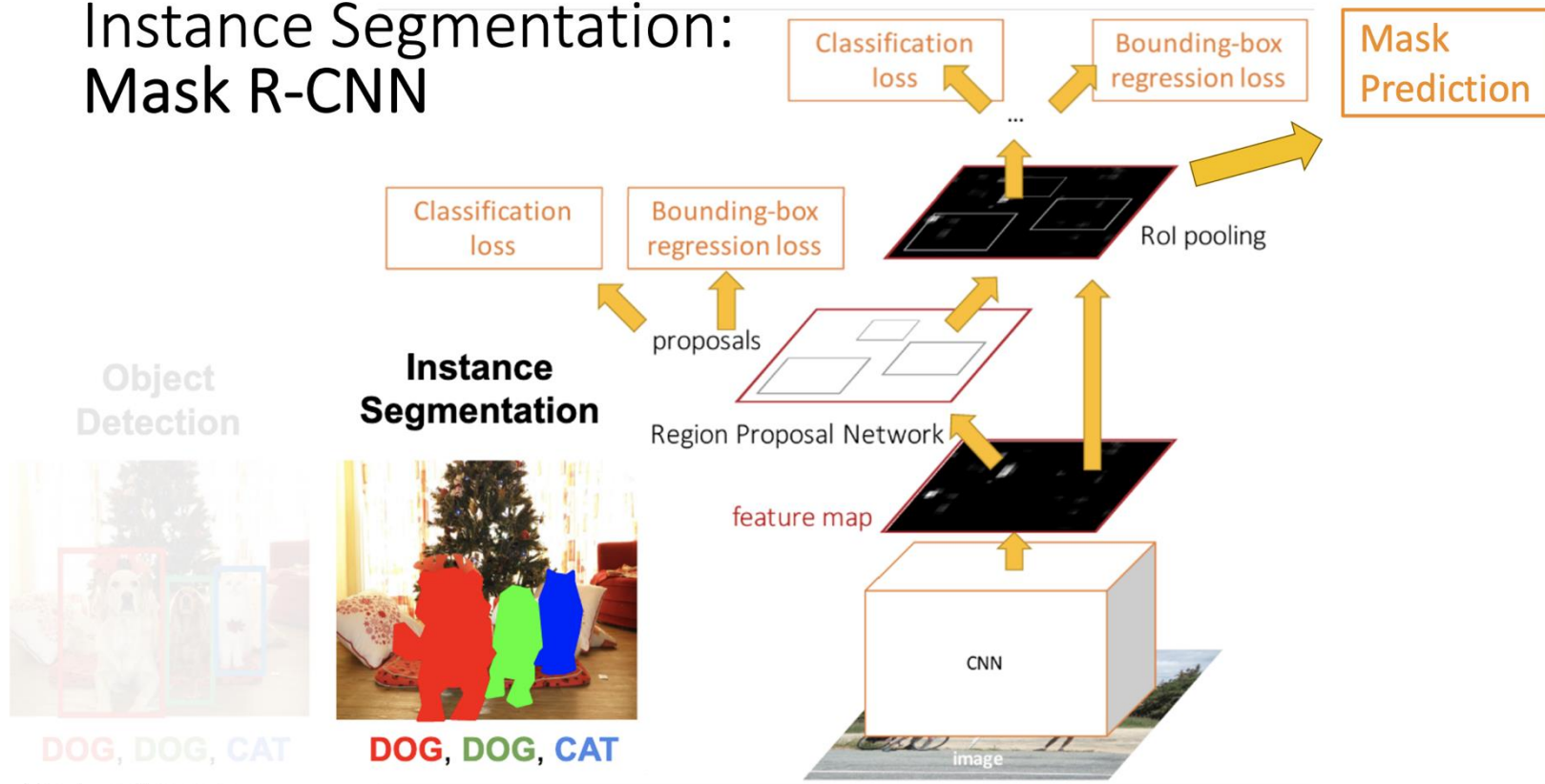
Object Detection: Faster R-CNN



Ren et al, "Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks", NeurIPS 2015

INSTANCE SEGMENTATION

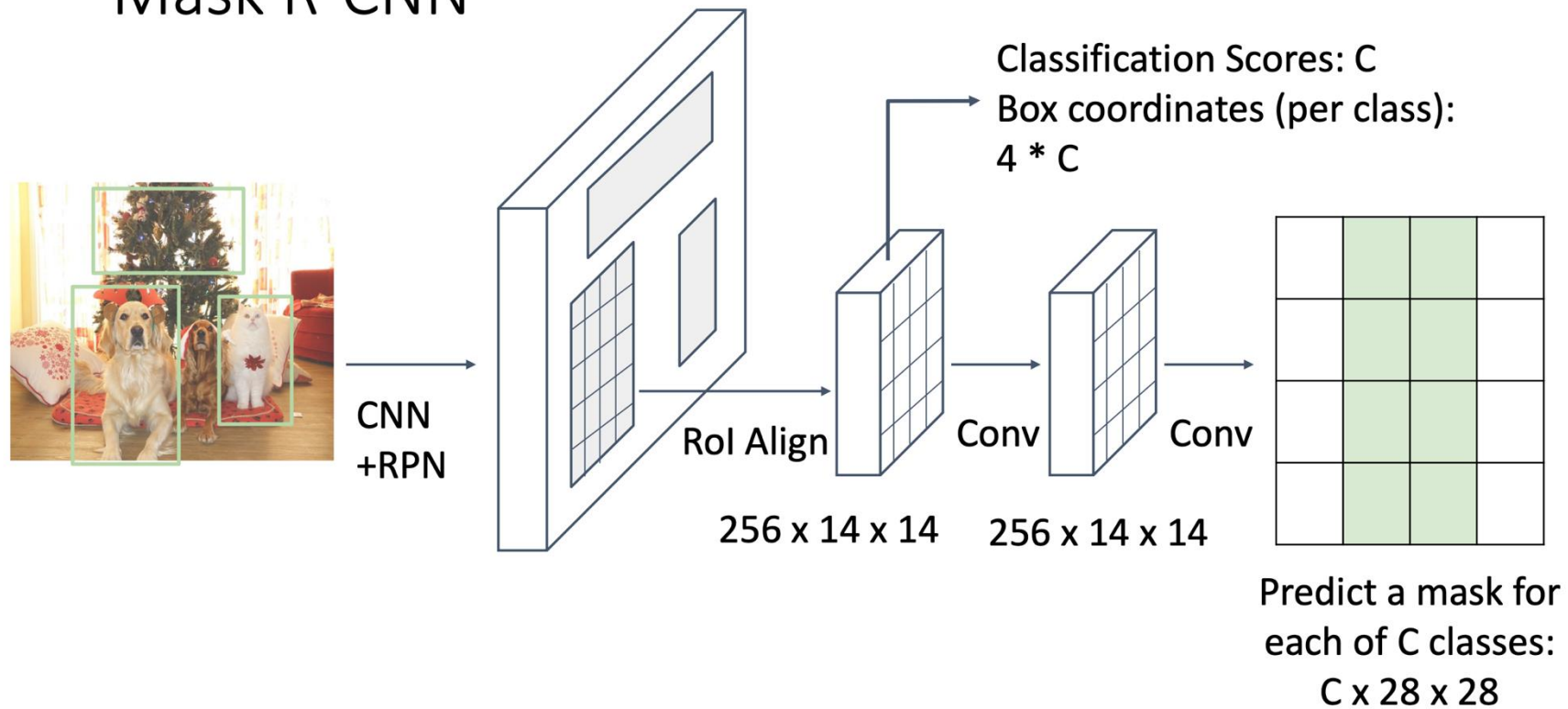
Instance Segmentation: Mask R-CNN



He et al, "Mask R-CNN", ICCV 2017

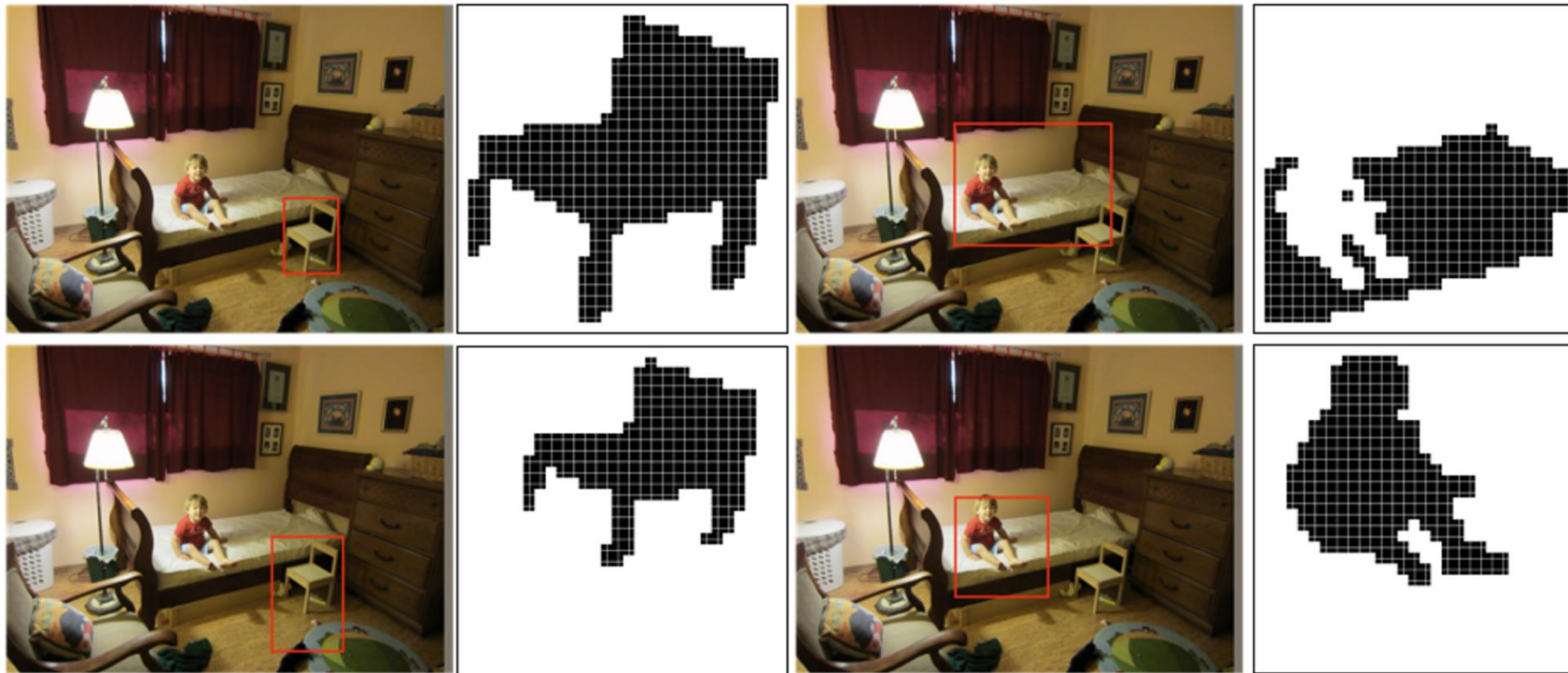
INSTANCE SEGMENTATION

Mask R-CNN



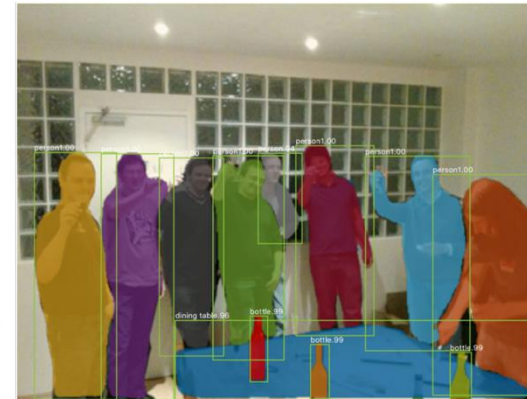
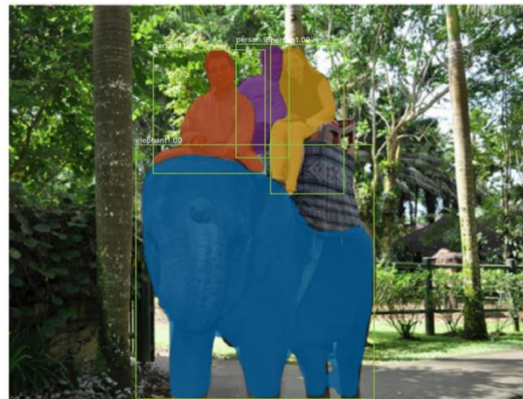
INSTANCE SEGMENTATION

Mask R-CNN: Example Training Targets



INSTANCE SEGMENTATION

Mask R-CNN: Very Good Results!



SUMMARY UNTIL NOW

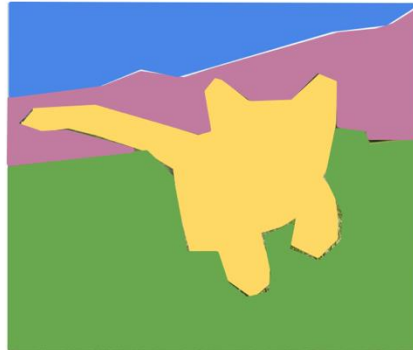
Classification



CAT

No spatial extent

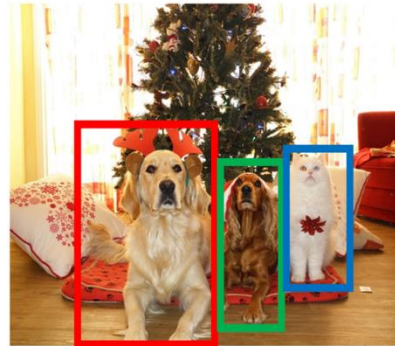
Semantic Segmentation



GRASS, CAT, TREE, SKY

No objects, just pixels

Object Detection



DOG, DOG, CAT

Instance Segmentation



DOG, DOG, CAT

Multiple Objects

[This image is CC0 public domain](#)



An abstract graphic on a light gray background. Two thin, dark gray lines intersect. One line is nearly vertical, starting from the top center and extending towards the bottom right. The other line is nearly horizontal, starting from the top left and extending towards the middle right. The word "SUMMARY" is written in a bold, black, sans-serif font, positioned to the right of the intersection of the two lines.

SUMMARY

SUMMARY

1. 2D Semantic / instance segmentation



NEXT LECTURE

3D Object detection



Two thin, dark gray lines intersect diagonally on a light gray background. One line runs from the top-left towards the bottom-right, and the other runs from the top-right towards the bottom-left. They intersect in the upper-left quadrant of the image.

QUESTIONS?